

eDNA MODE Build Guide:

Housing & Filtration Units

Control Unit & Sampling Unit

Part 1: Cutting and assembling PVC

Parts:

- 4 inch inner diameter Schedule 40 PVC
- PVC Glue
- PVC Primer
- PVC End Cap
- PVC Union

Tools:

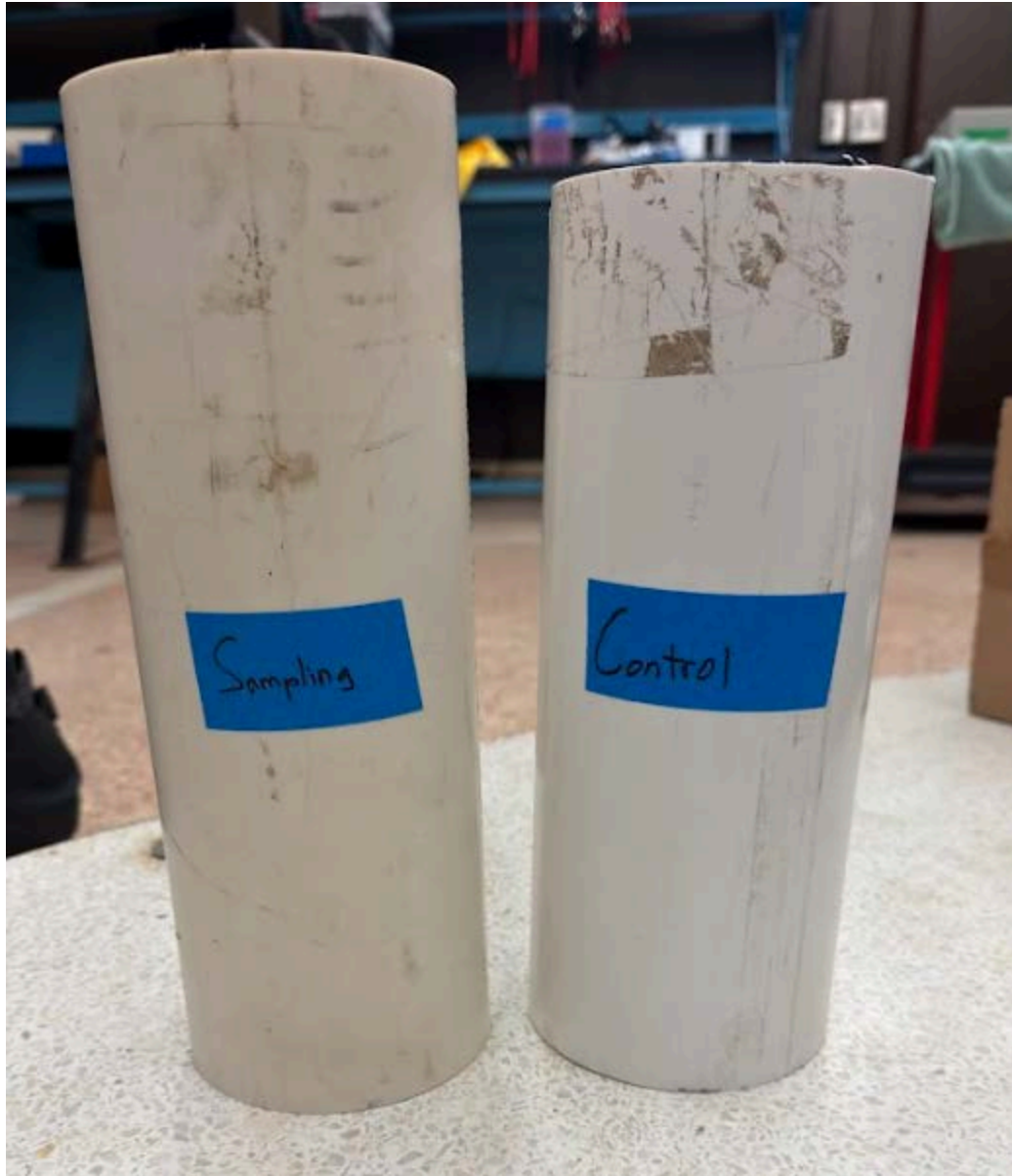
- Hand Saw
- Vice or Clamps



1. Mark PVC at 28 cm for control housing
2. Secure PVC for cutting



3. Begin cut at 90 degree angle
4. Cut through entire PVC cleanly



5. Repeat steps for sampling housing except length for sampling housing cut is 30 cm.



6. Use sandpaper to clean edges of PVC end
7. Wipe off PVC dust
8. WARNING - Complete **Section 3** of *Part 2* before moving on for ease of assembly



9. Coat Inside of cap with PVC Primer



10. Coat outside of PVC tube with primer

11. Immediately after, coat same area of cap and tube with PVC cement



12. Immediately after, insert cap onto PVC and press firmly on cap while rotating to ensure snug fit onto PVC pipe.
13. Press until the cap will no longer go any deeper
14. Repeat steps on other end of tube using PVC Union
15. Repeat all steps 6 - 13 for sampling housing

Part 2: Drilling Acrylic Disks for Sled and Penetrators

Parts:

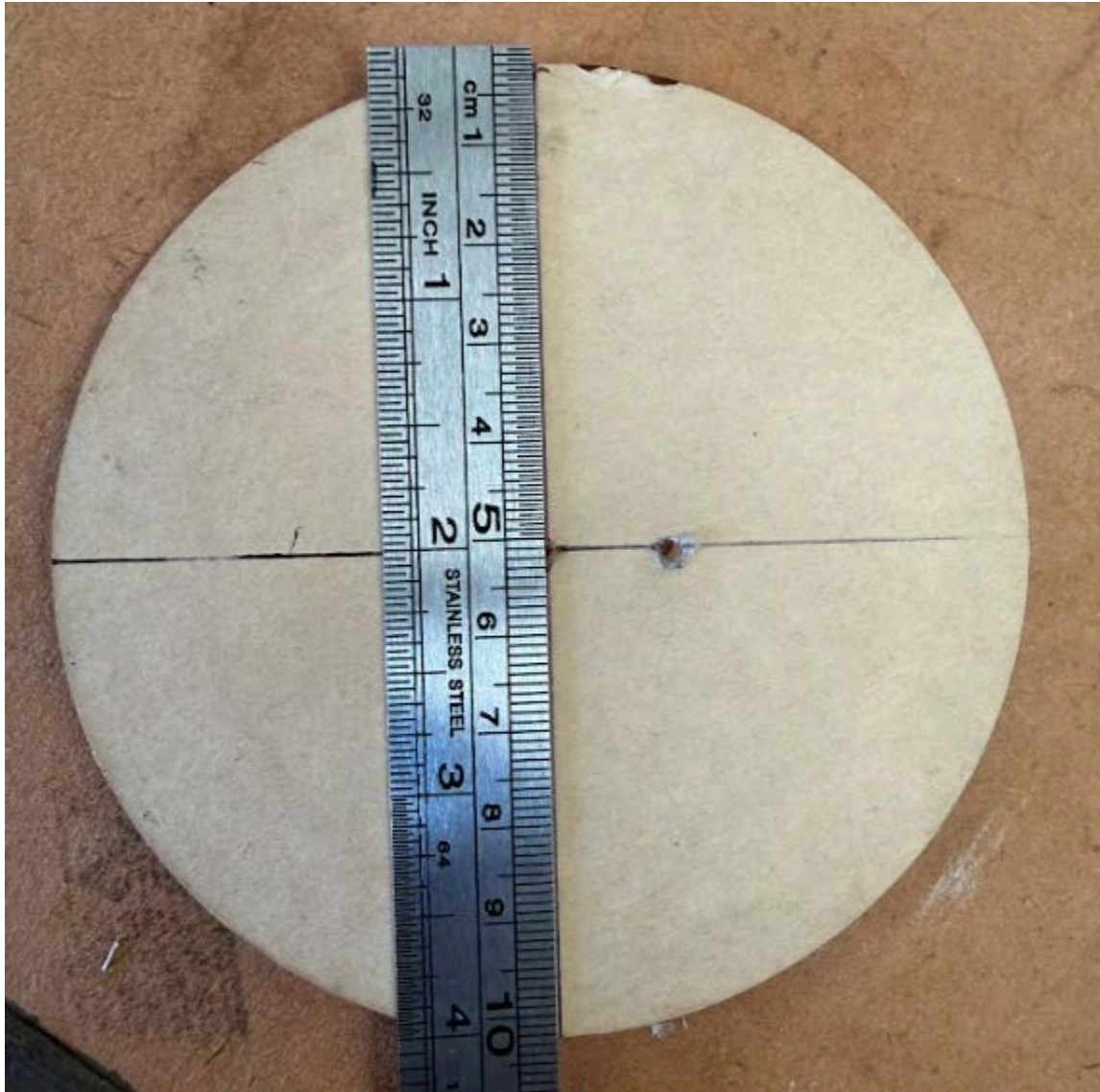
- 4 4-inch diameter acrylic disks
- 2 5 ¼ -inch diameter acrylic disks
- Mounting Brackets
- Acrylic Sleds (One 5 ½ x 2 inch section) (One 5 x 2 inch section) Both with Velcro adhesive attached.

Tools:

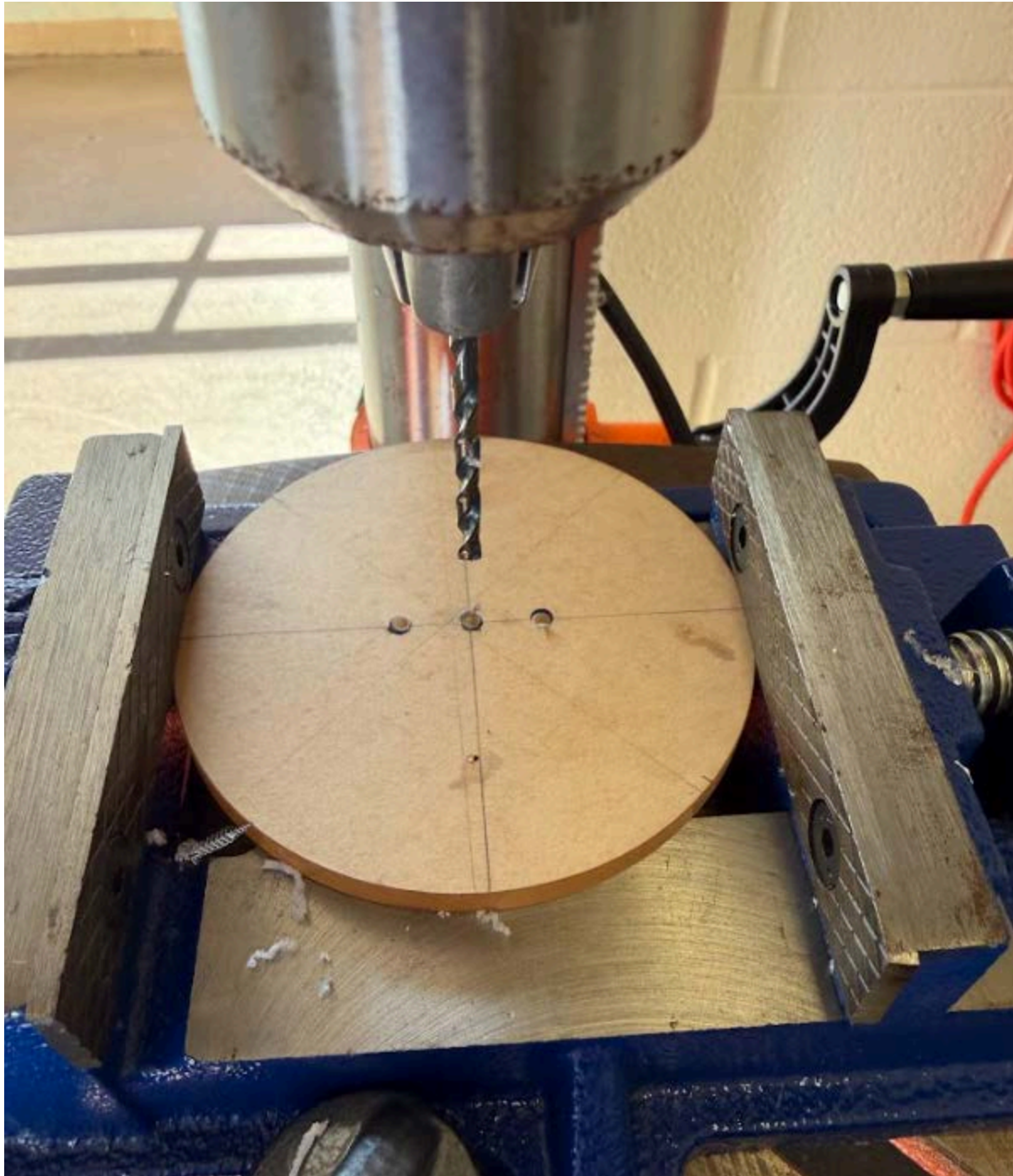
- M10 Drill bit
- M3 Drill Bit
- M13 Drill Bit
- Drill Press

- Drilling Oil
- Clamps
- Ryobi Belt and Disk Sander

Section 1: Inner Acrylic Disks



1. Grab 4 inch outer diameter acrylic disk
2. Find center and mark
3. Use Mounting brackets to mark other 2 holes for drilling using center as center of mounting bracket



1. Attach M3 Drill bit into Drill press
2. Secure disk with vice
3. Align holes to drill press
4. Lubricate drill bit and drilling area
5. Set Drill speed to 700 RPM
6. Drill holes through slowly



1. On same disk, mark vertical line halfway from top to middle and from bottom to middle
2. Drill pilot holes with M3 drill bit
3. Switch out drill bit to M13 drill bit
4. Drill over the just drilled M3 hole with the M13 bit. Leave mounting bracket holes alone



1. Create 2 of these designs with 3 holes for brackets and 2 for wire passages

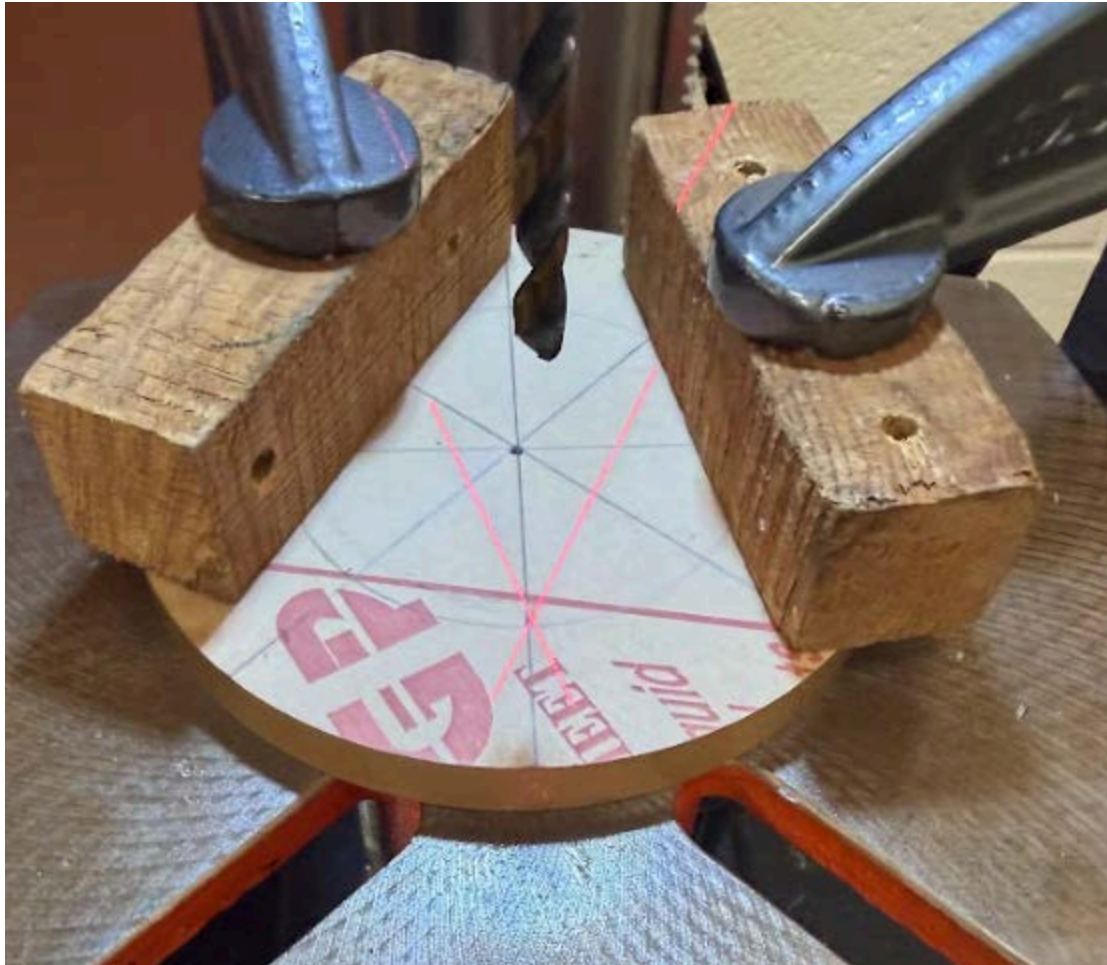


1. Create 1 of this design. Three holes for brackets, one hole for cable passage
2. Create 1 design with only 3 holes for brackets.
3. You should now have 4 4-inch acrylic disks
4. Using the Ryobi belt and disk sander, sand the around the edges of the disks evenly. Remove approximately 3 mm of acrylic material. This will ensure a clean fit into the PVC housings.

Section 2: PVC Pipe Union Acrylic Disks



1. Retrieve 5 ¼ inch diameter acrylic disks
2. This is for the 9 penetrator acrylic disk used in the sampling unit
3. Find center and mark
4. Mark into 8 equal sections
5. Set circle geometry tool to 4 mm and mark from center
6. You should have 8 points where circle and lines intersect and 1 point in the center
7. This totals 9 points where holes will be drilled for penetrators



8. Secure disk using C-Clamps and wooden block to prevent damage to disk
9. Install M3 drill bit into drill press
10. Repeat drilling procedure from 4 inch acrylic disks
11. Drill all 9 holes
12. Install M10 drill bit into drill press
13. Go back over all 9 holes with M10 drill bit



14. Final product should look as above



15. Grab another 5 ¼ inch acrylic disk for control unit
16. Create an equilateral triangle within an equilateral triangle from center
17. Mark points where triangles touch
18. You should have 3 equally spaced points
19. Install M3 drill bit and drill pilot holes for the points you marked
20. Switch to M10 drill bit and repeat drill press procedure



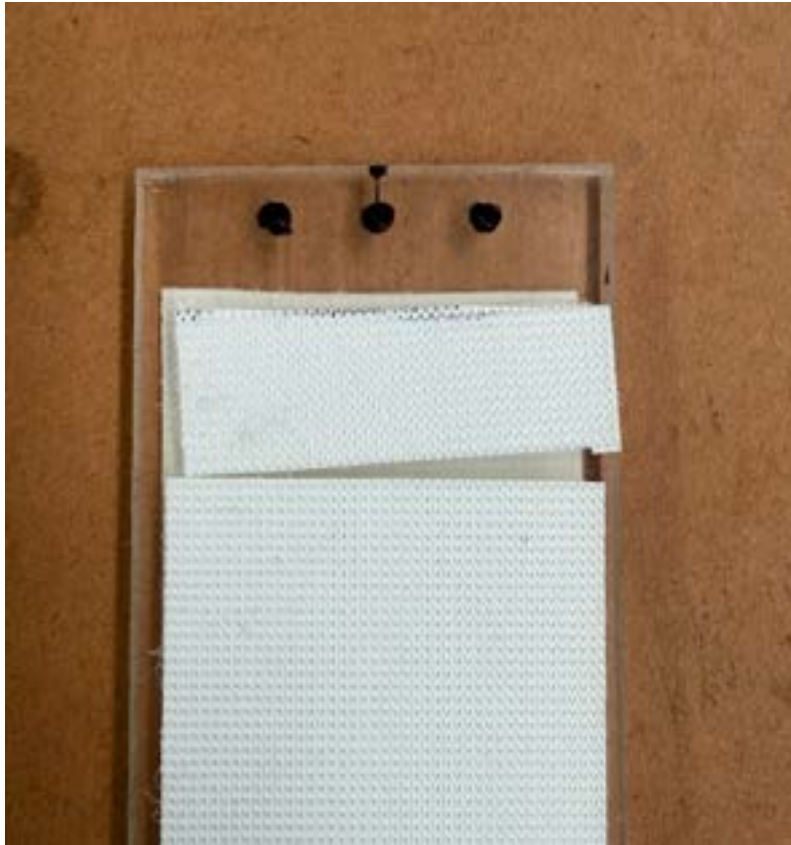
21. Final Product should resemble this

Section 3: PVC Pipe End Cap

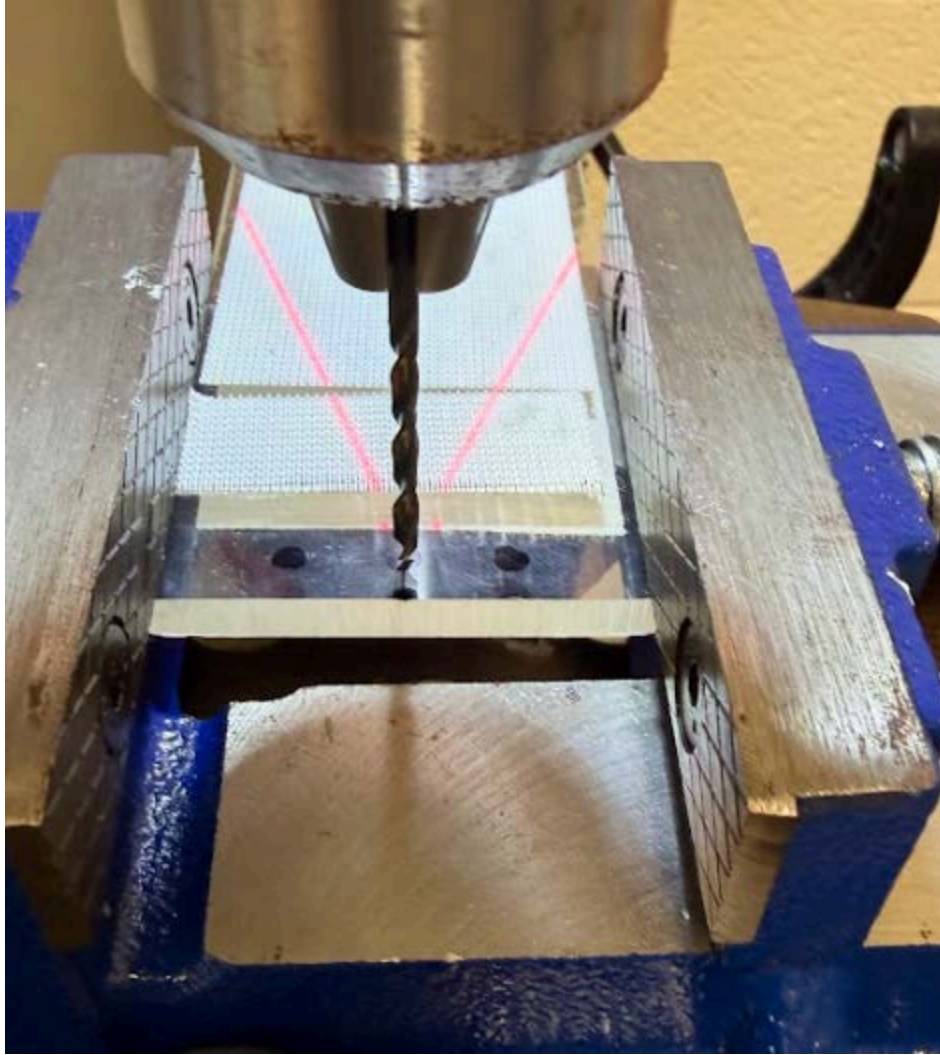


1. Get 1 PVC end cap for sampling unit
2. Secure in drill press
3. Drill with M10 near the center. Placement is not critical just near center
4. Product should resemble this
5. Other end cap does not need any modification

Section 4: Acrylic Sled Drilling.



1. Grab 5 ½ x 2 inch section of sled
2. Mark holes for brackets in middle of sled at the end
3. Repeat for both ends of the sled



4. Secure to drill press with vice
5. Install M3 drill bit
6. Repeat Drill press procedure
7. Drill all 6 holes in sled



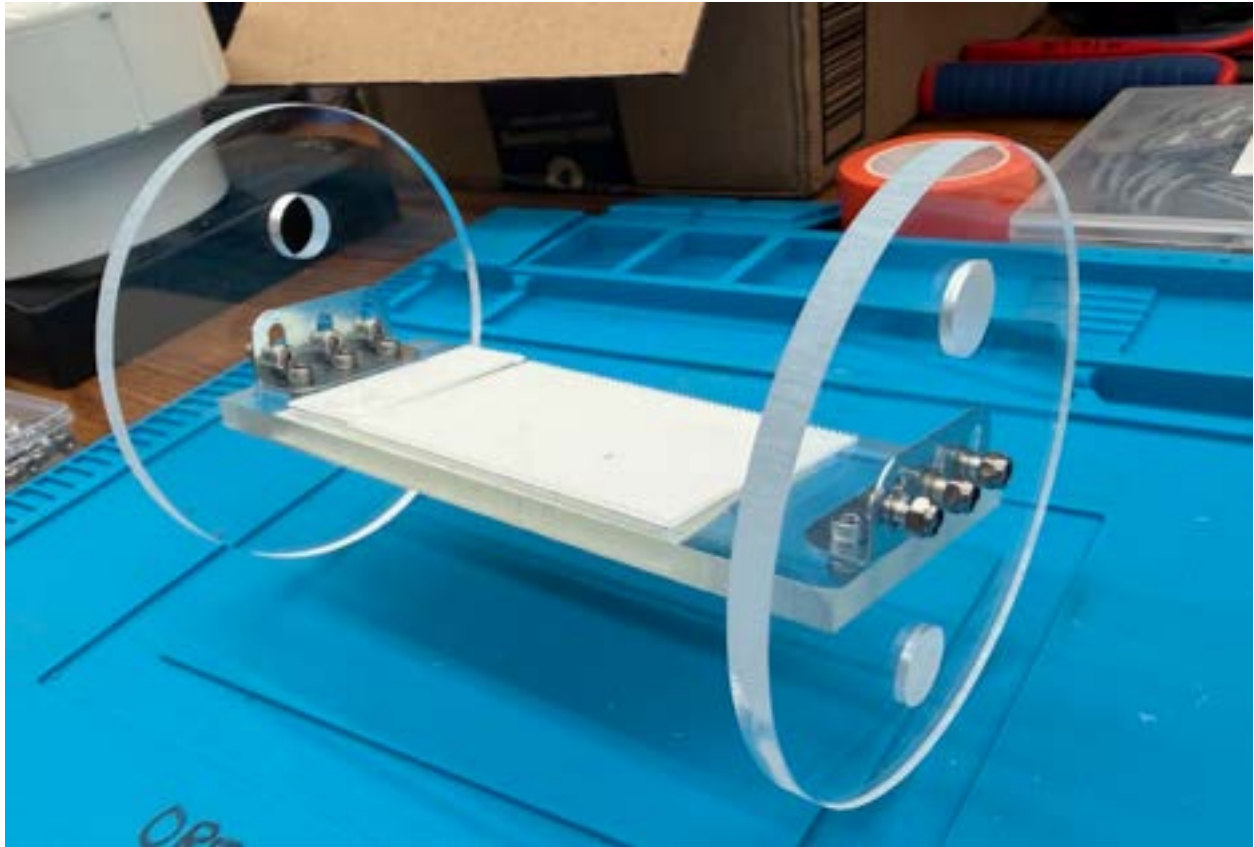
8. Grab 1 bracket, 3 2.5M bolts, and 3 2.5M lock nuts



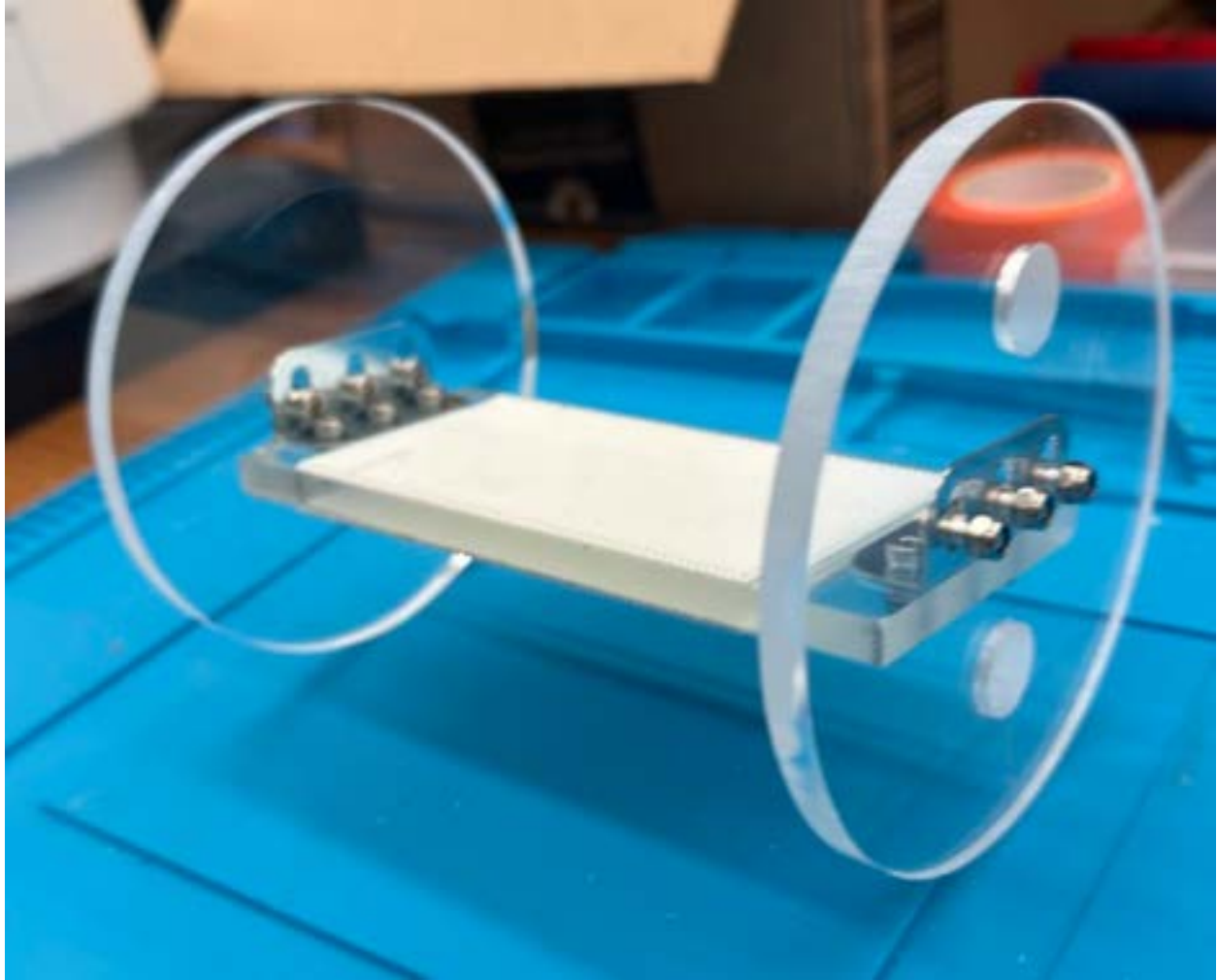
9. Attach bracket to end of acrylic sled



10. Attach the 2-cable hole acrylic disk to end



11. Attach the 1-cable hole acrylic disk to the other end
12. Finished Sampling unit sled should look like photo above



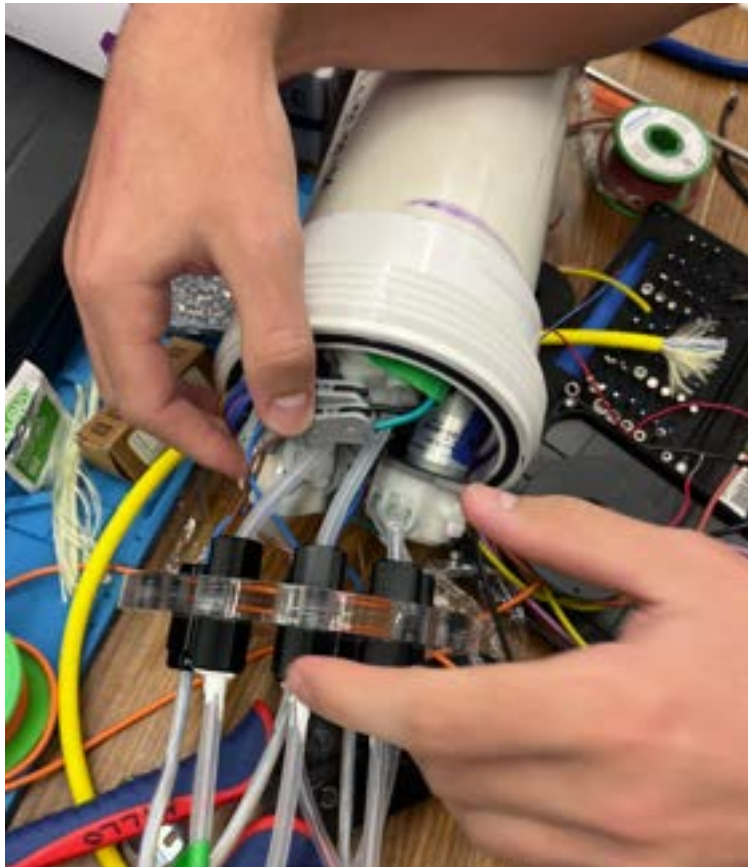
13. Repeat all section steps for 5 x 2 inch sled section
14. Attach 1 2-cable hole acrylic disk to end
15. Attach last remaining 0-cable hole acrylic disk to other end
16. Finished Control unit sled should look like photo above

Assembling Housings

Sampling Unit:

1. Attach a 7.5 HC penetrator to the one hole on the closed side of the sample unit. Strip 8 cm of yellow cable and thread the wires through the penetrator into the sample house.
2. Tighten the penetrator.
3. Attach velcro adhesive to all electrical components in sampling unit and attach to the sampling unit sled.

4. Thread wires from yellow cable through the side of the sled with a single hole. Making connections between these wires and the sampling unit components.
5. Wires connected to the preservative and sampling pumps should be routed through the holes on the 2-hole acrylic plate.
6. The sled should slide into the sampling unit with the 1-hole acrylic plate side entering first. The acrylic plate with 2 holes should be closest to the PVC union.
7. Attach correct connections to peristaltic pumps. Refer to the electrical build guide for these connections.
8. Zip tie extra wires together and push back inside the sampling unit to the acrylic plate.
9. Place peristaltic pumps inside the sampling unit with the tubing side closest to the PVC union. Ensure there are no kinks in the tubing as you place the 9 penetrator acrylic plate down, closing the sampling unit.



10. Screw on the PVC union.

Controls Unit:

1. Take the acrylic sled set up that has one plate with two holes and one plate with no holes. This is the control unit sled.
2. Place velcro adhesive onto all control unit electrical components. Attach these components to the sled.
3. Thread all wire connections through the two holes on one side of the sled.
4. Slide the sled into the brain housing with the blank acrylic plate side sliding in first. Attach all wire connections to the yellow cable wires, LED light and switch.

5. Place this acrylic plate down and tighten the union over the acrylic plate. Now the control unit and sampling unit are connected and sealed.



Assembling the 3 Filtration Units

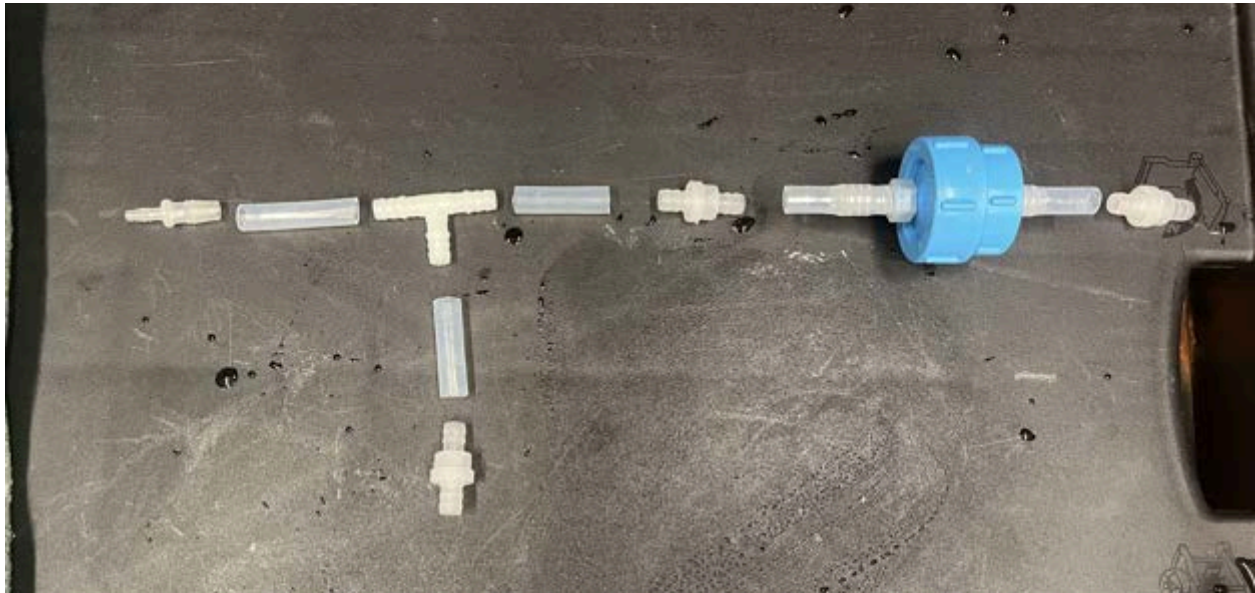
Parts:

- 3x Advantec Polypropylene Filter Holder
- 6x hose barb fitting
- 3x barb reducer
- 9x check valves
- 3x T connects
- 25 ft of 3/8" ID Silicon Tubing

- Zip ties

Tools:

- Utility knife / blade
- Surface to cut on
- Small pliers



1. Cut 15x 5 cm lengths of the $\frac{3}{8}$ in. ID tubing.
2. Put together your filter cartridge system by taking one hose barb fitting and screwing it onto the top of the filter cartridge. Take the other hose barb fitting and screw it onto the bottom of the filter cartridge.
3. Lay out your parts as pictured in the orientation in the picture.
4. Starting from the top left of the picture, begin by connecting one hose barb reducer to a piece of 5 cm length $\frac{3}{8}$ in. ID tubing. Then, connect this to one side of the top of the T fitting.
5. Connect 2 other pieces of 5 cm length $\frac{3}{8}$ in. ID tubing to the other side and bottom of the T fitting.
6. At the other end of the tubing connected to the bottom of the T fitting, connect a check valve in the orientation as shown. There is an arrow on the check valve that should point towards the T fitting if it is oriented correctly.
7. At the end of the tubing connected to the right of the T fitting, connect a check valve. The check valve should be oriented so that the arrow points away from the T fitting.
8. Connect the filter cartridge system to this check valve. The check valve arrow should be pointed at the top of the filter cartridge.
9. At the other end of the filter cartridge system, connect a check valve. The check valve arrow should be pointed away from the filter.

10. Tie zip ties around all check valve connections with tubing. This should add up to 6 zip ties per filtration unit.



11. Use scissors to cut the ends of the zip ties as close to the “knot” as possible



12. Repeat these steps for all 3 filtration units.

Preparing your Pumps

Preparing the Sampling Pumps

Parts:

- 3x Comet 12 V submersible pumps

- 3x prefilters

Tools:

- gloves
- Marine epoxy putty
- Sharpie
- Step drill bit
- Drill
- Sandpaper
- Alcohol wipes
- Paper towels



1. With the step drill bit and a hand held drill, drill a hole in the center of the bottom of the prefilter that is roughly the radius of the comet pumps.
 - a. Do this by tracing the comet pumps on the bottom of the prefilter with a silver sharpie or by stopping drilling and periodically checking if the pump fits for each step.
 - b. Do this step 3x for all prefilters.



2. Place each comet pump into a prefilter and push the pump all the way until it hits the top of the prefilter.
3. Trace a line around the comet pump with a black sharpie where the bottom of the prefilter ends. This is where the bottom “screw cap” of the prefilter will be epoxied to the comet pump.



4. Sandpaper the comet pumps half an inch above and below this line. Use an alcohol wipe to remove the plastic debris after using the sandpaper. Then, use a paper towel to dry off the pumps.



5. Using marine epoxy putty, fill in gaps between the comet pump and the “cap” of the prefilter. Only pack in marine epoxy putty on the side of the cap closest to the pump wires. Ensure the prefilter “cap” sits perpendicular the the pump while the epoxy starts to harden.



6. Let the pumps dry overnight before submerging them in water again.

Replace Preservative Pump Tubing

1. Gather your 3 preservative pumps.
2. Open all pumps by squeezing the white tabs on the side of the pump and removing the “caps.” Inside you should see a yellow triangle squeezing white flexible tubing.



3. Remove the triangle and the flexible tubing.
4. Cut 3x 40 cm 4 mm ID tubing.
5. Replace the flexible tubing with this tubing being sure to remove the green tabs from the flexible tubing and placing them on this new tubing. Make sure that there is an equal amount of tubing exiting from both sides of the pump.
6. Place the yellow triangle back in place. This will require a little maneuvering to fit the tubing and triangle.



7. If you have a hard time replacing the tubing, use a popsicle stick to push the tubing past the yellow triangle and into place.
8. Repeat these steps for each of the three peristaltic pumps.

Preparing your Sampling Unit Acrylic Plate

Part 3: Potting the Penetrators for Preservative Tubing

Parts:

- Acrylic disk with 9 holes
- 6x Potted Cable Penetrator (M10 for 8mm cable)
- 6x M10 8mm cable potted penetrators
- 3x 5.5mm HC penetrators
- 2x 7.5 HC penetrators
- 1x White LED penetrator
- 1x Switch Penetrator

Tools:

- Marine epoxy
- 4x Syringes
- Alcohol wipes
- Chem wipes
- Plastic cups
- Popsicle sticks
- Gloves
- Sharpie
- Drying stand

1. Prepare penetrators to be screwed into the 9-hole acrylic plate.
 - a. Wipe down o-rings with alcohol wipes followed by chem wipes.
 - b. Wipe down o-ring groove in penetration with alcohol wipes followed by chem wipes.
 - c. Grease o-ring with marine grease. Inspect o-ring for any hair, dust, and debris and remove if you see any.
 - d. Place the o-ring into the groove in the penetrator
 - e. Screw the penetrator into the plate
 - f. Use this guide from Blue Robotics for the [assembly and installation of penetrators](#).
2. Screw 6x potted cable penetrators into holes as oriented in the picture below.



3. Use a stand to hold the acrylic plate with penetrators and the three preservative pumps below it.
4. Orient the pumps so that the motor is positioned toward the center of the disk.



5. For each pump, thread the inlet and outlet tubing into penetrators that are side by side.
6. Each pump will have a different length of inlet and outlet tubing sticking out on the other side of the acrylic plate.
 - a. For pump 1, 9.5 cm of tubing should stick out on the other side of the acrylic disk for both inlet and outlet tubing.
 - b. For pump 2, 11.5 cm of tubing should stick out.
 - c. For pump 3, 13.5 cm of tubing should stick out.
 - d. This allows the pumps to sit staggered within the sample housing.
7. Pot penetrators using [directions for potting penetrators](#) given by Blue Robotics as a guide.



8. Let the epoxy dry fully for 24 hours.

Thread Sampling Pump Cables Through the Penetrators

1. Mark a line on the cable 50 cm and 60 cm from the sample pump for all three sample pumps.



2. Snip the wire at 60 cm. Strip the wire until the 50 cm mark for all 3 water pump cables. Do this using small wire cutters and snip carefully around the sheath in a circle to avoid nicking the inner wires. Use [BlueRobotics instructions](#).
3. After stripping you should see two wires, one blue and one brown.
4. Strip both blue and brown wires about 2 cm down using a wire stripper. Use 1.6 MM or 14 AWG gauge for stripping wires.



5. Each cable can now be inserted into the three open penetrators in the acrylic disk using [Blue Robotics instructions](#). Tightening these penetrators is shown below.



6. For sampling unit union acrylic disk, repeat penetrator installation steps with 3x 5.5mm HC penetrators
7. For control unit union acrylic disk, repeat penetrator installation steps with 1x 7.5mm HC penetrator, 1x white LED penetrator, and 1x switch penetrator
8. For PVC end cap for sampling unit, repeat penetrator installation steps using 1x 7.5mm HC penetrator

Altering the Pelican Case

Drill Holes for Sampling Pumps in the Pelican Case

1. Connect the step drill bit to the hand held drill.



2. Trace the three prefilters on one side of the pelican case with a sharpie. Make sure they are at the bottom of the case but as close to the opening as possible. See below for the prefilter positions and labeling.
 - a. The center of each prefilter circle should be about 8 cm away from the next circle.



3. Drill holes about the same radius as these traced pre filters but periodically check as you are drilling if the hole is the right size by slipping the prefilters into the holes. It should be a tight fit.



4. Prepare marine epoxy by pulling the epoxy apart with your fingers.



5. Place all three prefilter tops into the holes, Only the threads of the prefilters should be peaking through.

6. Use putty marine epoxy to seal the tops of the prefilters to the case on the outside.



7. The finished product should look like the image below.



Drill Outlet Hole in Pelican Case

1. On the end of the pelican case opposite from the sampling pump prefilters, mark a point at the same height and half of the width of the case.
2. Attach the following drill bit to the hand held drill.



3. Drill into the pelican case with an 11/16 inch flat boring bit at the point you marked.



4. Sand the hole to remove any sharp edges from drilling into the plastic.



5. Attempt to fit the cable gland for outflow through the hole. If it does not fit, sand more until it does fit.



6. Insert a length of $\frac{3}{8}$ in ID tubing through this cable gland. Cut the tubing about 2 cm extended from the cable gland on the outside of the pelican case and about 5 cm extended from the cable gland on the inside of the pelican case.

Integrating Sampling Units into the Pelican Case

Filtration Unit - Full Assembly with Sampling Unit

1. Use 6 barbed connectors and connect them to all inlet and outlet preservative tubing sticking out of the sampling unit acrylic plate.
2. Mark inlet and outlet for preservative tubing, along with the pump number the tubing corresponds to. (We use green tape for outlet, purple tape for inlet (syringe))
3. Cut 3x 4 cm of 4 mm ID tubing.



4. Connect 3x 4 cm of the 4 mm ID tubing to the inlet tubing for each pump.
5. Connect 3x syringe barbs to 3x 10 mL syringes by screwing the barbs onto the tip of the syringe.
6. Connect 3x 10 mL syringes to these connections
7. Cut 3 more lengths of 4 mm ID tubing for the outlet preservative tubing lines.
 - a. Pump 1: 6.5 cm
 - b. Pump 2: 10 cm
 - c. Pump 3: 19 cm
8. Connect this to the outlet preservative tubing lines sticking out of the acrylic plate making sure to match the lengths with the correct pump number.
9. Cut 3 different lengths of $\frac{3}{8}$ in ID tubing for the inlet water pump lines.
 - a. Pump 1: 10 cm
 - b. Pump 2: 14 cm
 - c. Pump 3: 12 cm
10. Connect this tubing to the corresponding water pump and a filtration unit. Once you connect the tubing to a filtration unit label that unit pump 1,2, or 3 using tape and a sharpie. Sample pumps should be oriented as pump 1 at the bottom of this picture and pump 3 at the top of this picture.



11. Connect your outlet preservative tubing lines to the filtration unit matching your preservative pump number.



12.

13. Cut 3x 5 cm length $\frac{3}{8}$ in ID tubing

14. Connect these to the check valve at the bottom of the filtration unit and then to the 4-connect as pictured below.



15. Connect the end of the 4-connect to the tubing sticking out of the cable gland.

16. Your filtration units should be connected as seen below.



17. Finally, zip tie all tubing connections (e.g., between check valves and tubing, between barbs and tubing, between filters and tubing)

Preparing eDNA MODE for Deployment

Preparing Preservative Reservoir

1. Disconnect the preservative outlet tubing end from the barb reducer connected to the filtration unit for all filtration systems.
2. Disconnect the preservative inlet tubing from the syringes for all filtration systems. Fill all syringes with 7 mL of preservative and put to the side.
3. Fill a small cup with 30 mL of preservative solution.
4. Schedule all preservative pumps to run for 10 seconds using the eDNA MODE user control panel by setting sample duration to 0 seconds and preservative pump runtime for 10 seconds. Interval can be set to zero.
5. Place all ends of the preservative tubing into this solution and hold the tubing.
6. Allow the preservative pumps to run and fill all tubing.
7. After all lines are filled, redo the connections of the preservative tubing to the filtration systems and the syringes.